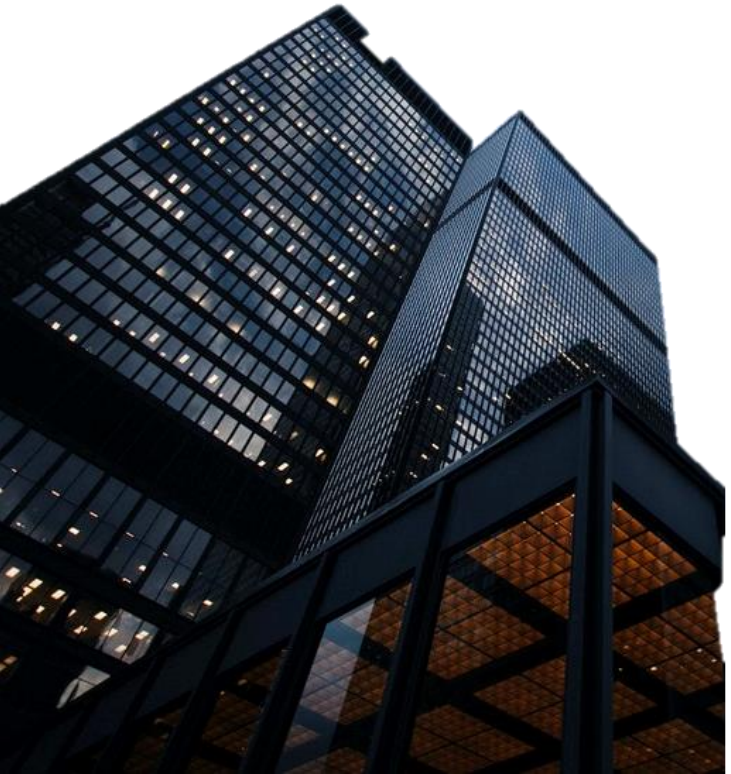
CE 208
Quantity Surveying

Estimation of RCC Beams and Columns



DEPARTMENT OF CIVIL ENGINEERING
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

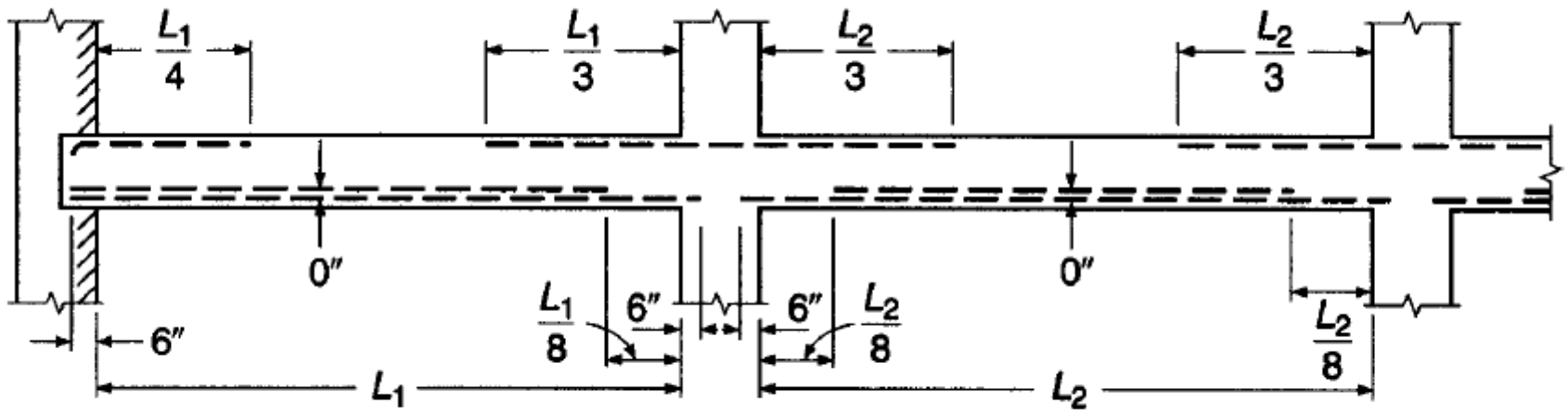
Reinforcement – Usual Sizes

- Slab – No. 3, 4, 5 (10mm, 12mm, 16mm)
- Beam – No. 5, 6, 7, 8 (16mm, 20mm, 22mm, 25mm)
- Stirrup/Tie – No. 3, 4 (10mm, 12mm)
- Column – No. 5, 6, 7, 8, 9, 10, 11, 14, 18
- Mat – No. 4, 5, 6, 8 (12mm, 16mm, 20mm, 25 mm)
- **Smaller sizes preferred as long as there is no congestion**

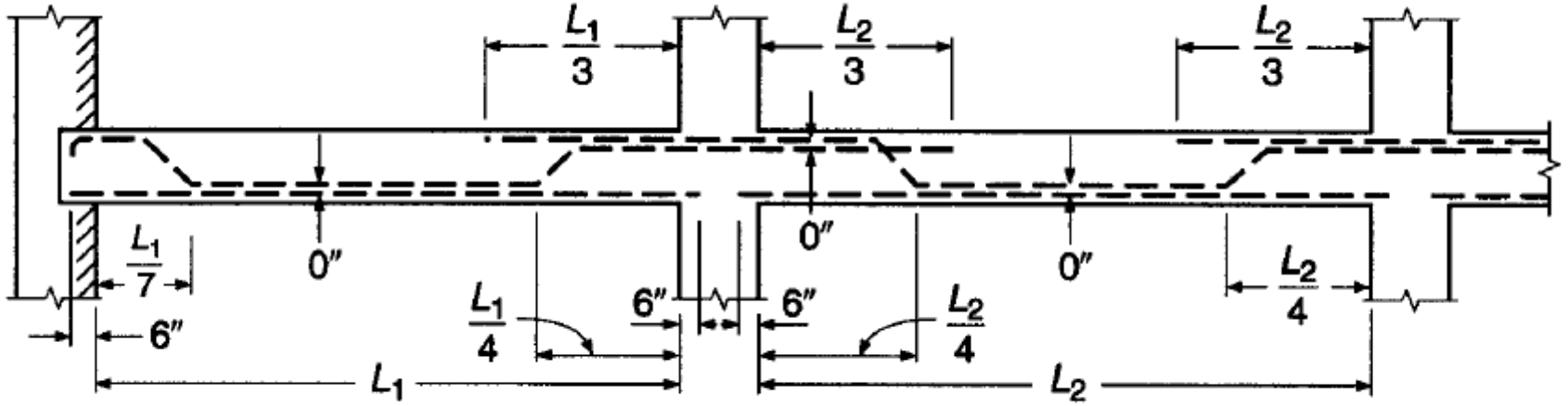
Main Reinforcement (Beams)

- Reinforcement ratio – 0.003 to 0.02 (Varies with material properties) (0.3% to 2%)
- Minimum 4 longitudinal bars when rectangular stirrups are used
- Longitudinal reinforcements are often cut-off or cranked according to requirements
- Stirrup spacing varies from zone to zone
- Bend length – 4 to 10 times d_b (Varies with bar diameter) – in general, assume $6d_b$

Cut-off and Bend points in Beams

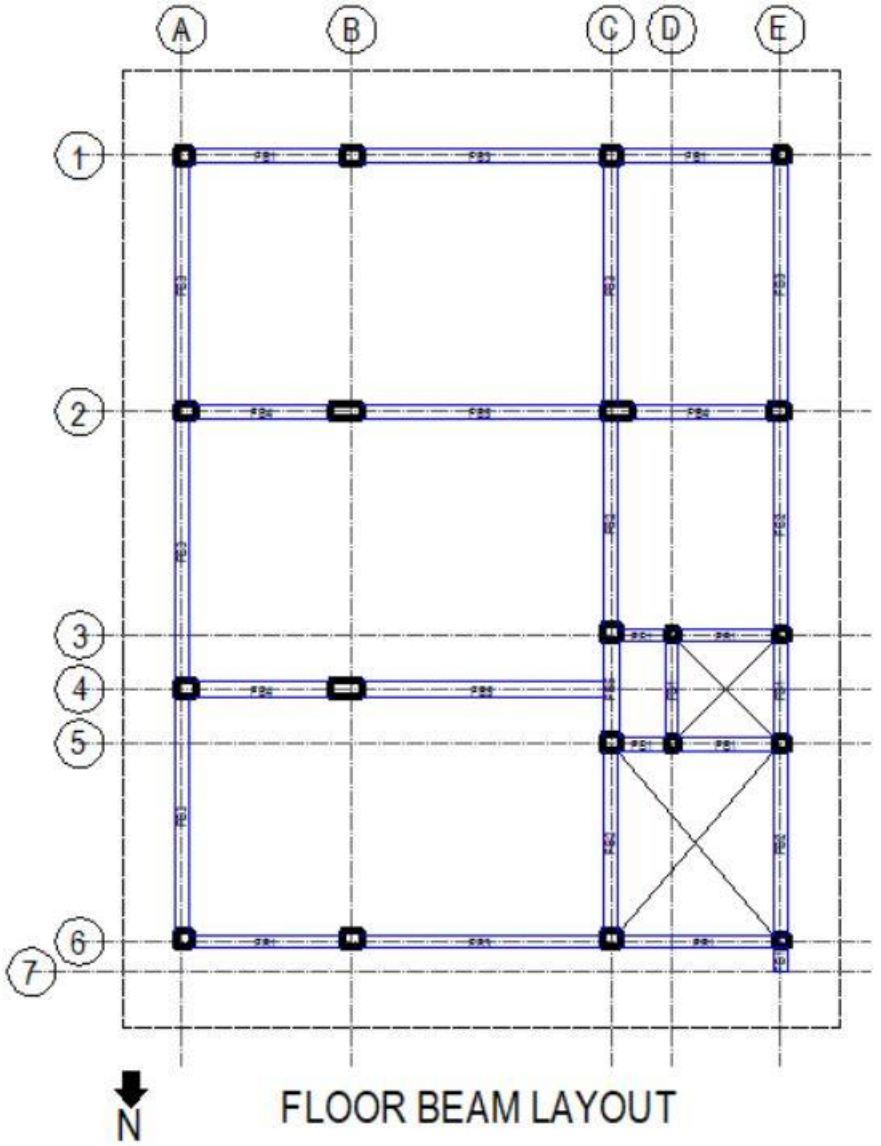


(a)

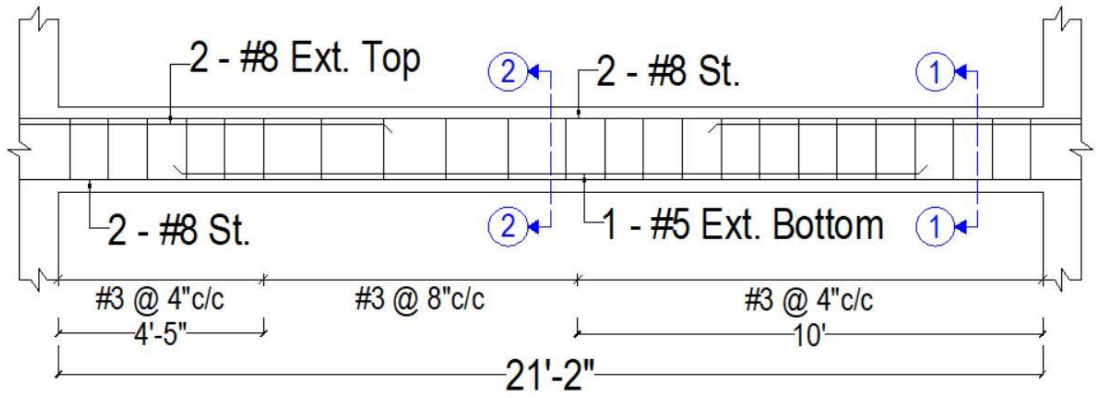


(b)

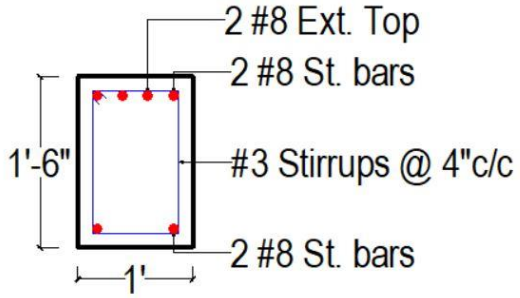
Beam Layout Plan



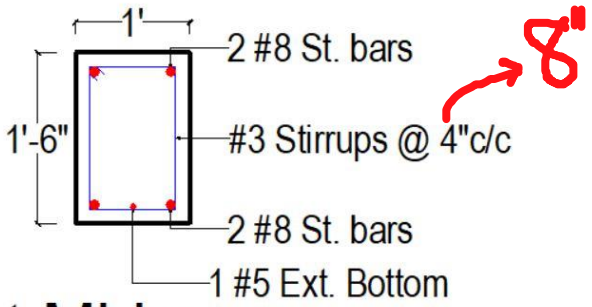
Beam Sections



Longitudinal Section of Floor Beam



At support



At Midspan

SECTION OF FB5

Estimation of Beams

Estimate the total amount of materials required for FB5 Beams of the six storied building drawings provided.

Estimation of Beams

ESTIMATION OF BEAMS

Item No.	Item	Bar Designation	No.	Bar Dia (ft)	Item Multiplier	Straight Length (ft)	Hooked Length (ft)	Total Length (ft)	Area (sft)	Volume (cft)
3	Floor Beam Reinforcement	FB5								
		Straight Bar	24	0.083333	4	22.167	1.000	92.667	0.005486	12.2011
		Extra Top	24	0.083333	4	7.556	1.000	34.222	0.005486	4.5059
		Extra Bottom	24	0.052083	1	15.875	0.625	16.500	0.002083	0.8250
		Stirrups	24	0.031250	64	56 hobe 4.0000	0.375	280.000	0.000764	5.1333

Unit weight of mild steel (in kg/cft): 222.323049

Volume of #8 dia bar (in cft):	16.707	Weight of #8 dia bar (in kg):	3714.3594139
Volume of #5 dia bar (in cft):	0.825	Weight of #5 dia bar (in kg):	183.4165154
Volume of #3 dia bar (in cft):	5.133	Weight of #3 dia bar (in kg):	1141.2583182

Costing of 60 grade mild steel per kg: 80

Total #8 dia bar cost (in BDT):	297149
Total #5 dia bar cost (in BDT):	14674
Total #3 dia bar cost (in BDT):	91301
Total Reinforcement Cost (in BDT):	403124

Item No.	Item	Description	No.	Length (ft)	Width (ft)	Height/ Thickness(ft)	Area (sft)	Volume (cft)	Total Volume (dry) (cft)	Total Volume (wet) (cft)
4	Floor Beam Concrete (1:2:4)	FB5	24	21.17	1.00	1.50	x	762.00	762.00	1143.00

Costing of brick chips per cft:	110	Total volume of brick chips (in cft):	653.14
Costing of sand per cft:	30	Total volume of sand (in cft):	326.57
Costing of cement per bag:	500	Total quantity of cement (in bags):	131
Total Brick Chips Cost (in BDT):	71846		
Total Sand Cost (in BDT):	9798		
Total Cement Cost (in BDT):	65500		
Total Concrete Cost (in BDT):	147144		

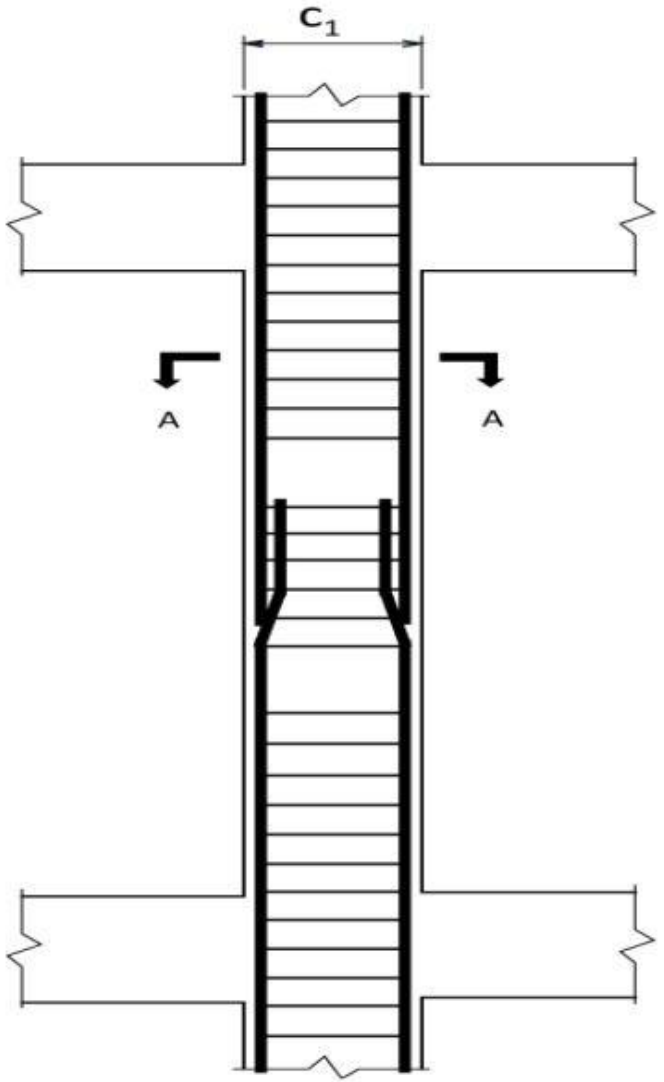
TOTAL COST OF BEAMS (IN BDT): 550268

Main Reinforcement (Columns)

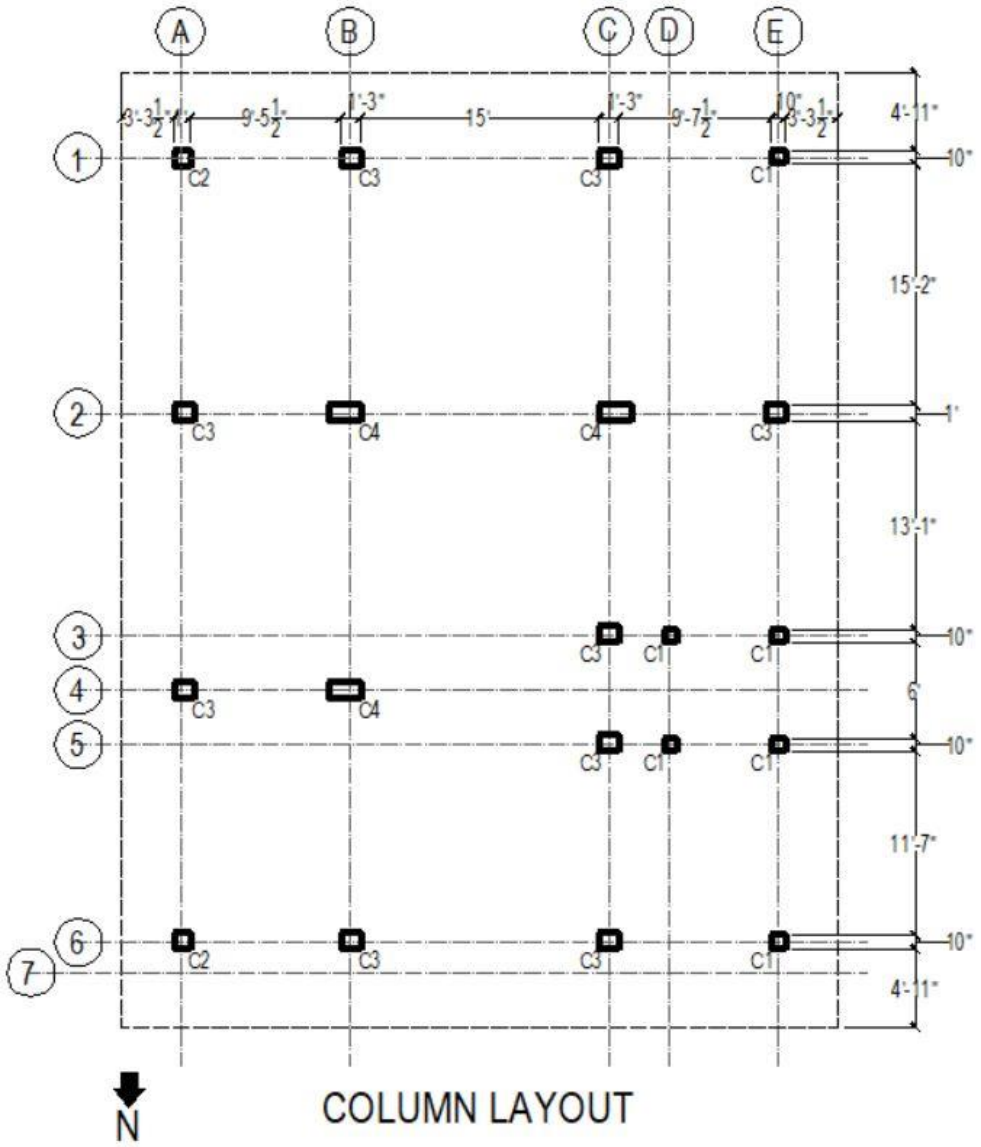
- Reinforcement ratio – 0.01 to 0.08 (1% to 8%)
- Normally 2-3 %, preferably not exceed 4%.
- Minimum 4 longitudinal bars when rectangular or circular ties are used
- Minimum 6 bars when continuous spiral is used
- Bend length – 4 to 10 times d_b (Varies with bar diameter) – in general, assume $6d_b$
- Lap Length – Must be greater than 300mm (12 inches)

Column Lap/Splices

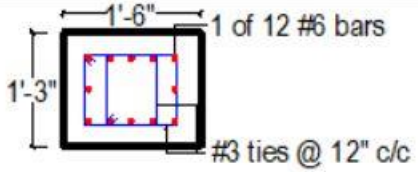
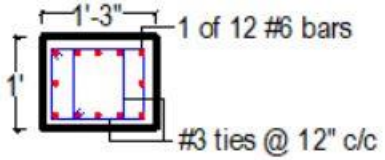
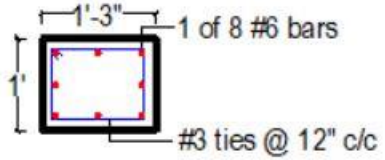
- Maximum 50% of the main bars can be lapped in a section
- Lap Length – Must be greater than 300mm (12 inches)
- Lap must be away from the joint
- Lap location depends on the length of reinforcements



Column Layout Plan



Column Sections

LOCATION BETWEEN	COLUMN SECTION
BASE TO GF	
GF TO 3F	
3F TO ROOF	

COLUMN SCHEDULE (C3)

Estimation of Columns

Estimate the total amount of materials required for C3 Columns of the six storied building drawings provided.

Estimation of Columns

ESTIMATION OF COLUMNS

Item No.	Item	Bar Designation	No.	Bar Dia (ft)	Item Multiplier	Straight Length (ft)	Hooked/Lap Length (ft)	Total Length (ft)	Area (sft)	Volume (cft)
1	Column Reinforcement (Above PL)	C3 (Vertical Reinforcement)								
		PL to 3rd floor	9	0.062500	12	30.00	1.00	372.00	0.003056	10.2300
		3rd floor to roof	9	0.062500	8	30.00	1.00	248.00	0.003056	6.8200
		C3 (Horizontal Reinforcement)								
		PL to 3rd floor	9	0.031250	60	6.06	0.75	408.75	0.000764	2.8102
		3rd floor to roof	9	0.031250	60	3.50	0.38	232.50	0.000764	1.5984

Unit weight of mild steel (in kg/cft): 222.323049

Volume of #6 dia bar (in cft):	17.050	Weight of #6 dia bar (in kg):	3790.6079855
Volume of #3 dia bar (in cft):	4.409	Weight of #3 dia bar (in kg):	980.1320043

Costing of 60 grade mild steel per kg: 80

Total #6 dia bar cost (in BDT):	303249
Total #3 dia bar cost (in BDT):	78411
Total Reinforcement Cost (in BDT):	381660

Item No.	Item	Description	No.	Length (ft)	Width (ft)	Height/ Thickness(ft)	Area (sft)	Volume (cft)	Total Volume (dry) (cft)	Total Volume (wet) (cft)
2	Column Concrete (Above PL) (1:2:4)	Above PL							675.00	1012.500
		Column C3	9	1.25	1	60	x	675.00		

Costing of brick chips per cft:	110	Total volume of brick chips (in cft):	578.57
Costing of sand per cft:	30	Total volume of sand (in cft):	289.29
Costing of cement per bag:	500	Total quantity of cement (in bags):	116
Total Brick Chips Cost (in BDT):	63643		
Total Sand Cost (in BDT):	8679		
Total Cement Cost (in BDT):	58000		
Total Concrete Cost (in BDT):	130322		

TOTAL COST OF COLUMNS (IN BDT): 511982

Assignment

Estimate the total amount of materials required for C4 Columns and FB3 Beams of the six storied building drawings provided.

Thank You!